Modeling of Vaccine Tender Problem

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1 Two-Stage Stochastic Program Mathematical Formulation - Max Profit

1.1 Deterministic Equivalent Model

This model represents the maximization of profit for the producers.

1.1.1 Model Notation

Sets:

- $\mathcal{V}$: set of vaccines (from the WHO pre-qualified vaccine list),
- $\mathcal{V}_a$: subset of vaccines containing antigen a,
- $\mathcal{A}$: set of antigens,
- $\mathcal{A}_v$: subset of antigens in vaccine v,
- $\mathcal{A}_p$: subset of antigens contained in producer p's portfolio,
- $\mathcal{P}$: set of producers,
- $\mathcal{P}_v$: subset of producers of vaccine v,
- $\mathcal{T}$: set of time periods in the planning horizon,
- $\mathcal{T}'_t$: subset of periods when a tender ends (since we allow tenders of at most Δ years, $\mathcal{T}'_t = \{t, t+1, ..., min\{t+\Delta, |\mathcal{T}|\}\}$.
- Ω: set of scenarios for uncertain parameters

Parameters:

- $\tilde{d}_{at\omega}$: stochastic demand at time t for antigen a in scenario ω,
- $\tilde{s}_{pt\omega}$: stochastic production capacity of producer p at time t in scenario ω,
- $\bar{s}_p$: base production capacity of producer p,
- r_{vpt}: the reservation price of vaccine v produced by p in period t,
- $\bar{r}_{vt}$: average price of vaccine v across all producers in period t,
- l_{vp}: the annualized return on investment required by producer p for vaccine v,
- g_t: set-up cost for initiating a tender in period t,
- f_{vpt}: the production set-up cost of producer p for vaccine v in period t,
- h_v: the annual holding cost for vaccine v as a proportion of its price,

- β: penalty per unit of shortage of the committed amount, i.e., the missed dose penalty,
- Γ_p: capacity extension cost for producer p,
- κ: allowable capacity extension rate for producer p,
- n: maximum value that capacity extension variable, L_{pt}, can get.
- δ_t: a discount factor to capture the time value of money
- Δ: maximum allowable tender length.
- $\hat{I}_{v0}$: initial inventory level for vaccine v,
- $\hat{S}_{a0}$: initial unvaccinated children condition of antigen a,
- $\hat{F}_{a0\tau}$: ongoing tenders at the beginning of the planning horizon for antigen a by the period τ,
- $\hat{Q}_{vp0\tau}$: commitment amount for the ongoing tender of vaccine v by producer p by the period τ.

Decision Variables:

- $F_{at\tau}$: a binary variable indicating whether a tender covers the demand for antigen a for periods t through τ,
- Y_{pt}: a binary variable which specifies whether or not manufacturer p produces in period t,
- $W_{pt\tau}$: a binary variable indicating if producer p has made commitments for periods t through τ.
- L_{pt}: an integer variable representing capacity extension for producer p in period t.
- $Q_{vpt\tau}$: procurement commitment for vaccine v by producer p for periods t through τ,
- $X_{vpt\omega}$: doses of vaccine v delivered by producer p in period t for scenario ω,
- $I_{vt\omega}$: stock level for vaccine v at the beginning of period t (including period 0) for scenario ω,
- $V_{vt\omega}$: number of doses administered with vaccine v in period t for scenario ω,
- $S_{at\omega}$: number of missed doses for antigen a in period t (including period 0) for scenario ω.

1.1.2 Model Formulation

The first stage of the two-stage stochastic program model formulation is given as NU:

$$\min - \sum_{t\in\mathcal{T}} \delta_t (\sum_{p\in\mathcal{P}} -\Gamma_p L_{pt}) + \mathbb{E}_\Omega[\mathcal{Q}(\Xi, \omega)] \tag{1}$$

$$(\tau - t + 1)F_{at\tau} \leqslant \sum_{p\in\mathcal{P}_a} \sum_{l=t}^{\tau} Y_{pl}, \qquad \forall a \in \mathcal{A}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{2}$$

$$F_{at\tau} + F_{at\tau'} \leqslant 1, \qquad \forall a \in \mathcal{A}; t \in \mathcal{T}; \tau, \tau' \in \mathcal{T}'_t; \tau \neq \tau'. \tag{3}$$

$$\sum_{t=max\{1,l-\Delta\}}^{l} \sum_{\tau\in\mathcal{T}'_t} F_{at\tau} \geqslant 1, \qquad \forall a \in \mathcal{A}; l \in T. \tag{4}$$

$$\sum_{a\in\mathcal{A}_p} F_{at\tau} \geqslant W_{pt\tau}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{5}$$

$$\sum_{a \in \mathcal{A}_p} F_{at\tau} \leqslant |\mathcal{A}_p| W_{pt\tau}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{6}$$

NU:

$$\hat{L}_{pt\tau} = \sum_{l=t+1}^{\tau} (\tau - l + 1)\kappa \bar{s}_p L_{pl}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{7}$$

$$\check{L}_{pt\tau} = \sum_{l=1}^{t} (\tau - t + 1)\kappa \bar{s}_p L_{pl}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{8}$$

$$\sum_{v \in \mathcal{V}_p} Q_{vpt\tau} \leqslant W_{pt\tau}(\sum_{l=t}^{\tau} \bar{s}_p) + \langle W_{pt\tau}\hat{L}_{pt\tau}\rangle^M + \langle W_{pt\tau}\check{L}_{pt\tau}\rangle^M, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{9}$$

$$F_{at\tau}, Y_{pt}, W_{pt\tau} \in \{0, 1\}, L_{pt} \in \{0, ..., n\}, Q_{vpt\tau} \geqslant 0 \tag{10}$$

The second stage of the two-stage stochastic program model is formulated as follows NU:

$$\mathcal{Q}(\overline{\Xi}, \omega) = \min - \sum_{t \in \mathcal{T}} \delta_t (\sum_{v \in \mathcal{V}} \sum_{p \in \mathcal{P}_v} r_{vpt} X_{vpt\omega} + \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} \eta R_{pt\omega}) \tag{11}$$

$$\grave{X}_{vpt\tau\omega} = \sum_{l=t}^{\tau} X_{vpl\omega}, \qquad \forall v \in \mathcal{V}; p \in \mathcal{P}_v; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{12}$$

$$Q_{vpt\tau} \geqslant \langle W_{pt\tau}\grave{X}_{vpt\tau\omega}\rangle^M, \qquad \forall v \in \mathcal{V}; p \in \mathcal{P}_v; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{13}$$

$$\ddot{L}_{pt} = \sum_{l=1}^{t} \kappa \bar{s}_p L_{pl}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}. \tag{14}$$

$$\sum_{v \in \mathcal{V}_p} X_{vpt\omega} \leqslant Y_{pt}\tilde{s}_{pt\omega} + \langle Y_{pt}\ddot{L}_{pt}\rangle^M, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}, \tag{15}$$

$$I_{vt-1,\omega} + \sum_{p \in \mathcal{P}_v} X_{vpt\omega} = V_{vt\omega} + I_{vt\omega}, \qquad \forall v \in \mathcal{V}; t \in \mathcal{T}. \tag{16}$$

$$\tilde{d}_{at\omega} - \sum_{v \in \mathcal{V}_a} V_{vt\omega} + S_{at-1,\omega} \leqslant S_{at\omega}, \qquad \forall a \in \mathcal{A}; t \in \mathcal{T}. \tag{17}$$

NU:

$$\sum_{v \in \mathcal{V}_p} r_{vpt} X_{vpt\omega} + R_{pt\omega} \geqslant \sum_{v \in \mathcal{V}_p} (1 + l_{vp}) f_{vpt} Y_{pt}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}; \omega \in \Omega. \tag{18}$$

$$I_{v0\omega} = \hat{I}_{v0}, \qquad \forall v \in \mathcal{V}. \tag{19}$$

$$X_{vpt\omega}, K_{vpt\tau\omega}, I_{vt\omega}, S_{at\omega}, V_{vt\omega} \geqslant 0 \tag{20}$$

1.2 Application of L-Shaped Method

If the SP identifies any infeasibilities, it produces feasibility cuts in equation (21), which are added to the MP to eliminate infeasible first-stage solutions in future iterations. If the SP is feasible, and the problem is not optimal, optimality cuts in equation (22) are generated. The optimality cuts are derived from the dual variables of the SP and provide a linear approximation of the second-stage cost function within the feasible region of the problem [1]. Note that MP and SP are solved iteratively until the L-shaped method converges to the optimal solution. The lower bound (LB) is obtained from the solution of the Master Problem (MP). The LB represents the best-known estimate of the objective function value considering the first-stage decisions and the linear approximations of the second-stage costs. This bound improves iteratively as more cuts are added. The upper bound (UB) is determined by evaluating the current first-stage solution and solving the corresponding second-stage SPs. The UB provides the actual objective function value for the current first-stage decision, incorporating the true second-stage costs under different scenarios. The UB is updated whenever a new feasible solution with a lower cost is found.

The following model presents the Master Problem (MP): NU:

$$MP: \quad \mathcal{Z}^M = \min -(\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} -\Gamma_p L_{pt} + \theta)$$

constraints (2) - (10)

$$\sum_{\omega\in\Omega} p_\omega(h_\omega - T_\omega \Xi)\pi^l_\omega \leqslant 0 \qquad \forall l \in \check{\mathcal{L}}. \tag{21}$$

$$\theta \geqslant \sum_{\omega\in\Omega} p_\omega(h_\omega - T_\omega \Xi)\pi^l_\omega \qquad \forall l \in \widehat{\mathcal{L}}. \tag{22}$$

In the MP formulation, p_ω represents the probability associated with each scenario $\omega \in \Omega$. Each scenario represents a realization of demand, $\tilde{d}_{at\omega}$ and a realization of production capacity, $\tilde{s}_{pt\omega}$. Sets $\check{\mathcal{L}}$ and $\widehat{\mathcal{L}}$ represent the set of feasibility and optimality cuts added to the MP, respectively. Note that Ξ denotes the decision variables, $\{F, Y, W, L, Q, \hat{L}, \check{L}, \ddot{L}\}$, defined in the MP, and the $\overline{\Xi}$ represents the values of these variables, $\{\overline{F}, \overline{Y}, \overline{W}, \overline{L}, \overline{Q}, \overline{\hat{L}}, \overline{\check{L}}\,\overline{\ddot{L}}\}$, determined after solving the MP. Given the $\overline{\Xi}$, model SP$(\overline{\Xi}, \omega)$ is a linear program. π^l_ω represents the extreme rays of the dual variables when SP$(\overline{\Xi}, \omega)$ is infeasible, and π^l_ω denotes the optimal dual variables when the SP$(\overline{\Xi}, \omega)$ is feasible. h^l_ω defines the right-hand-side of constraints in SP$(\overline{\Xi}, \omega)$. T_ω represents the technology matrix.

To overcome obtaining infeasibility in the SPs and to eliminate the feasibility cuts, we add an auxiliary variable ($R_{pt\omega}$) on the LHS of constraint (18). We also penalize this variable in the objective function of SPs. Updated constraint (18) together with the updated SP objective function, equation (23), prevent infeasibilities and exclusively generates optimality cuts. This ensures a feasible solution space throughout iterations, facilitating the optimization process for more efficient convergence and higher solution quality.

For a given $\overline{\Xi}$ and for each realization $\omega \in \Omega$ of the random problem parameters $\tilde{d}_{at\omega}$ and $\tilde{s}_{pt\omega}$, the following represents the subproblem SP$(\overline{\Xi}, \omega)$.

NU:

$$SP(\overline{\Xi}, \omega): \quad \mathcal{Q}(\overline{\Xi}, \omega) = \min -\sum_{t\in\mathcal{T}} \delta_t(\sum_{v\in\mathcal{V}}\sum_{p\in\mathcal{P}_v} r_{vpt}X_{vpt\omega} + \sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} \eta R_{pt\omega}) \tag{23}$$

constraints (12) - (20)

NU:

$$\sum_{v\in\mathcal{V}_p} r_{vpt}X_{vpt\omega} + R_{pt\omega} \geqslant \sum_{v\in\mathcal{V}_p} (1 + l_{vp}) f_{vpt}\overline{Y}_{pt}, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}. \tag{24}$$

$$\overline{Q}_{vpt\tau} \geqslant \langle \overline{W}_{pt\tau}\dot{X}_{vpt\tau\omega}\rangle^M, \qquad \forall v \in \mathcal{V}; p \in \mathcal{P}_v; t \in \mathcal{T}; \tau \in \mathcal{T}'_t. \tag{25}$$

$$\sum_{v \in \mathcal{V}_p} X_{vpt\omega} \leqslant \overline{Y}_{pt} \tilde{s}_{pt\omega} + \langle \overline{Y}_{pt} \ddot{L}_{pt} \rangle^M, \qquad \forall p \in \mathcal{P}; t \in \mathcal{T}, \tag{26}$$

We make the following updates in the objective function of the MP and the optimality cuts to employ multi-cut generation:

NU:

$$MP: \quad \mathcal{Z}^M = \min - \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} -\Gamma_p L_{pt} + \sum_{\omega \in \Omega} p_\omega \theta_\omega)$$

$$\theta_\omega \geqslant p_\omega (h_\omega - T_\omega \Xi) \pi_\omega^l \qquad \forall l \in \mathcal{L}, \omega \in \Omega. \tag{27}$$

References

[1] J. R. Birge and F. Louveaux, *Introduction to stochastic programming.* Springer Science & Business Media, 2011.